

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

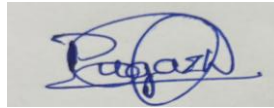
Total Marks:40

Code of Conduct:

I K.Pugazhendi/ 192311026 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANSWERS FOR ABOVE QUESTIONS:

Case Study 1 — Smart Medical Diagnosis System

(a) Representation using Propositional Logic

Let the following propositional symbols represent the patient's conditions and diagnoses:

F = Patient A has Fever

C = Patient A has Cough

B = Patient A has Breathlessness

R = Patient A has Rash

Flu = Patient A has Possible Flu

Measles = Patient A has Possible Measles

Pneumonia = Patient A has Possible Pneumonia

Rules

Rule I: Fever AND Cough $\rightarrow$ Possible Flu

$F \wedge C \rightarrow \text{Flu}$

Rule II: Fever AND Rash $\rightarrow$ Possible Measles

$F \wedge R \rightarrow \text{Measles}$

Rule III: Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

$C \wedge B \rightarrow \text{Pneumonia}$

Facts about Patient A

Given:

F=True

C=True

B=True

There is no fact stating that Patient A has Rash, so:

R=False/Unknown

Therefore, the knowledge base can be represented as:

$\text{KB} = \{F, C, B, F \wedge C \rightarrow \text{Flu}, F \wedge R \rightarrow \text{Measles}, C \wedge B \rightarrow \text{Pneumonia}\}$

(b) Forward Chaining

Forward chaining starts with the known facts and repeatedly applies rules whose conditions are satisfied to derive new facts.

Initial Facts

F=True

C=True

B=True

Step 1: Apply Rule I

Rule I:

$F \wedge C \rightarrow \text{Flu}$

We know:

Fever = True

Cough = True

$F \wedge C = \text{True}$

Hence:

Flu=True

Derived fact: Patient A has Possible Flu.

Step 2: Apply Rule II

Rule II:

$F \wedge R \rightarrow \text{Measles}$

We know:

Fever = True

Rash = not given

Therefore, the rule cannot be fired.

Measles cannot be derived

Step 3: Apply Rule III

Rule III:

$C \wedge B \rightarrow \text{Pneumonia}$

We know:

Cough = True

Breathlessness = True

Therefore:

$C \wedge B = \text{True}$

Hence:

Pneumonia = True

Derived fact: Patient A has Possible Pneumonia.

Forward Chaining Summary

Step	Rule	Conditions	New Fact
Initial	—	Fever, Cough, Breathlessness	Given facts
1	Rule I	$\text{Fever} \wedge \text{Cough}$	Possible Flu
2	Rule II	$\text{Fever} \wedge \text{Rash}$	Cannot fire
3	Rule III	$\text{Cough} \wedge \text{Breathlessness}$	Possible Pneumonia

Therefore, all possible diagnoses that can be derived are:

Possible Flu

Possible Pneumonia

Possible Measles cannot be derived because Rash is not known.

(c) Backward Chaining for Pneumonia

The goal is:

Pneumonia

Backward chaining starts with the goal and works backward to determine whether the required conditions can be satisfied by known facts.

Step 1: Start with the Goal

Goal = Pneumonia

We search for a rule that can conclude Pneumonia.

The relevant rule is:

$C \wedge B \rightarrow \text{Pneumonia}$

Step 2: Reduce the Goal

Pneumonia

becomes:

$C \wedge B$

So the goal reduction is:

Goal Pneumonia -- Rule III -- C and B

Step 3: Verify Cough

The first sub-goal is:

C

We already have the fact:

$C = \text{True}$

Therefore:

Cough is proved

Step 4: Verify Breathlessness

The second sub-goal is:

B

We already have the fact:

$B = \text{True}$

Therefore:

Breathlessness is proved

Complete Goal Reduction Tree

Goal Patient A has Pneumonia---Rule III:cough and Breathlessness--both given facts TRUE---goal satisfied---Pneumonia.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification of Rule 1 with the Given Facts

Rule 1

forall x: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

For applying unification, we consider the antecedent:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$

Given Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

We need to find a substitution for variable x that makes the rule conditions identical to the facts.

Step 1: Match $\text{Vehicle}(x)$

Rule condition:

$\text{Vehicle}(x)$

Fact:

$\text{Vehicle}(\text{Ambulance})$

Therefore:

θ

1

$=\{x/\text{Ambulance}\}$

After substitution:

$\text{Vehicle}(\text{Ambulance})$

This matches the given fact.

Step 2: Match $\text{EmergencyType}(x)$

Rule condition:

$\text{EmergencyType}(x)$

Fact:

$\text{EmergencyType}(\text{Ambulance})$

Using the substitution:

θ

1

$=\{x/\text{Ambulance}\}$

we get:

$\text{EmergencyType}(\text{Ambulance})$

which matches the given fact.

Therefore:

θ

2

$=\{x/\text{Ambulance}\}$

Complete MGU

The Most General Unifier (MGU) is:

$\theta = \{x/\text{Ambulance}\}$

Applying this substitution to Rule 1:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$

Since both antecedents are facts, we derive:

$\text{ClearPath}(\text{Ambulance})$

(b) Resolution Proof of $\text{Proceed}(\text{CarA})$

We want to prove:

$\text{Proceed}(\text{CarA})$

The relevant rules are Rule 2 and Rule 3.

Step 1: Convert Rules to CNF

Rule 1

Original:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

we get:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2

Original:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

This can be split into two clauses:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

and

$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3

Original:

$\text{Vehicle}(x) \wedge \text{Behind}(x, y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF form:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Step 2: Ground the Relevant Facts

Given:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

$\text{Vehicle}(\text{CarA})$

$\text{Behind}(\text{CarA}, \text{Ambulance})$

Step 3: Derive $\text{ClearPath}(\text{Ambulance})$

From Rule 1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Using:

$\theta = \{x/\text{Ambulance}\}$

we get:

$\neg \text{Vehicle}(\text{Ambulance}) \vee \neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

Since:

$\text{Vehicle}(\text{Ambulance})$

resolve with:

$\neg \text{Vehicle}(\text{Ambulance})$

to obtain:

$\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

Now resolve with:

$\text{EmergencyType}(\text{Ambulance})$

Therefore:

$\text{ClearPath}(\text{Ambulance})$

Step 4: Derive $\text{GreenSignal}(\text{Ambulance})$

Rule 2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Substitute:

$x = \text{Ambulance}$

giving:

$\neg \text{ClearPath}(\text{Ambulance}) \vee \text{GreenSignal}(\text{Ambulance})$

We already derived:

$\text{ClearPath}(\text{Ambulance})$

Resolution gives:

GreenSignal(Ambulance)

Step 5: Apply Rule 3

Rule 3:

$\sim \text{Vehicle}(x) \vee \sim \text{Behind}(x,y) \vee \sim \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Substitute:

$x=\text{CarA}, y=\text{Ambulance}$

We get:

$\sim \text{Vehicle}(\text{CarA}) \vee \sim \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \sim \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$

We have the facts:

Vehicle(CarA)

Behind(CarA, Ambulance)

GreenSignal(Ambulance)

Resolving successively:

Proceed(CarA)

Resolution Chain

The entire reasoning can be summarized as:

Vehicle(Ambulance) + EmergencyType(Ambulance) $\rightarrow$ Rule 1 $\rightarrow$ ClearPath(Ambulance) $\rightarrow$ Rule 2 $\rightarrow$ GreenSignal(Ambulance) $\rightarrow$ (Vehicle(CarA) and Behind(CarA, Ambulance)) $\rightarrow$ Rule 3

Final Result

Proceed(CarA) is TRUE

The resolution proof succeeds because all three required conditions of Rule 3 are satisfied.

(c) Handling Two Simultaneous Emergency Vehicles

Is the current knowledge base sufficient?

No, the current knowledge base is not fully sufficient for safely handling two simultaneous emergency vehicles.

The existing rules are designed to identify an emergency vehicle and give it a clear path:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

and then:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x)$

However, there is no rule describing what happens when two emergency vehicles require the same road/path or signal at the same time.

Example

Suppose a second emergency vehicle is introduced:

Vehicle(FireTruck)

EmergencyType(FireTruck)

The existing Rule 1 can derive:

ClearPath(FireTruck)

and Rule 2 can derive:

GreenSignal(FireTruck)

So the system could have:

GreenSignal(Ambulance) and GreenSignal(FireTruck) simultaneously.

But the KB does not specify:

Which emergency vehicle gets priority?

What happens if both vehicles approach the same intersection?

How conflicting green signals should be handled?

How paths should be separated?

What happens when both vehicles are behind/approaching the same vehicle?

How to prevent collisions?

Therefore, additional knowledge is required.

Case Study 3 — Agricultural Expert Reasoning System

(a) Conversion into Conjunctive Normal Form (CNF)

P1: SoilDry $\rightarrow$ IrrigationNeeded

Original:

SoilDry $\rightarrow$ IrrigationNeeded

Step 1: Eliminate implication

$A \rightarrow B \equiv \neg A \vee B$

Therefore:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

CNF:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Original:

$(\text{IrrigationNeeded} \wedge \text{CropWheat}) \rightarrow \text{ApplyDripMethod}$

Step 1: Eliminate implication

$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Step 2: Apply De Morgan's Law

$(\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat}) \vee \text{ApplyDripMethod}$

Therefore:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

This is already in CNF.

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Original:

$(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \rightarrow \text{CropAtRisk}$

Step 1: Eliminate implication

$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Step 2: Move $\neg$ inward using De Morgan's Law

$(\neg \neg \text{ApplyDripMethod} \vee \neg \text{CropWheat}) \vee \text{CropAtRisk}$

Since:

$\neg \neg \text{ApplyDripMethod} = \text{ApplyDripMethod}$

we get:

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts

Given:

SoilDry=True

CNF:

SoilDry

Given:

CropWheat=True

CNF:

CropWheat

Complete CNF Knowledge Base

C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

C4: SoilDry

C5: CropWheat

(b) Resolution Refutation to Prove ApplyDripMethod

Step 1: Goal

We want to prove:

ApplyDripMethod

For resolution refutation, negate the goal:

$\neg \text{ApplyDripMethod}$

Add it to the CNF knowledge base:

C6: $\sim \text{ApplyDripMethod}$

Step 2: Resolve C1 with C4

C1: $\sim \text{SoilDry} \vee \text{IrrigationNeeded}$

C4: SoilDry

Resolve SoilDry :

IrrigationNeeded

Call this:

C7: IrrigationNeeded

Step 3: Resolve C2 with C5

C2: $\sim \text{IrrigationNeeded} \vee \sim \text{CropWheat} \vee \text{ApplyDripMethod}$

C5: CropWheat

Resolve CropWheat :

$\sim \text{IrrigationNeeded} \vee \text{ApplyDripMethod}$

Call this:

C8: $\sim \text{IrrigationNeeded} \vee \text{ApplyDripMethod}$

Step 4: Resolve C8 with C7

C8: $\sim \text{IrrigationNeeded} \vee \text{ApplyDripMethod}$

C7: IrrigationNeeded

Resolve IrrigationNeeded :

ApplyDripMethod

Call this:

C9: ApplyDripMethod

Step 5: Resolve C9 with Negated Goal

C9: ApplyDripMethod

C6: $\sim \text{ApplyDripMethod}$

Resolving complementary literals gives:

The empty clause represents a contradiction.

Therefore:

$\text{KB} \models \text{ApplyDripMethod}$

Hence, ApplyDripMethod is logically proven.

Resolution Proof Tree

$\text{SoilDry} + \sim \text{SoilDry} \vee \text{IrrigationNeeded} \rightarrow \text{IrrigationNeeded} + \sim \text{IrrigationNeeded} \vee \sim \text{CropWheat} \vee$
 $\text{ApplyDripMethod} + \text{CropWheat} \rightarrow \text{ApplyDripMethod} + \sim \text{ApplyDripMethod}$

Thus:

ApplyDripMethod

(c) Removing the Fact SoilDry

Suppose:

SoilDry

is removed from the KB.

The remaining facts are:

CropWheat

The rules remain:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

$\sim \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step 1: Can IrrigationNeeded be derived?

Rule P1 is:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

But there is no longer a fact:

SoilDry

Therefore:

IrrigationNeeded cannot be derived

Step 2: Can ApplyDripMethod be derived?

P2 requires:

$\text{IrrigationNeeded} \wedge \text{CropWheat}$

Although:

$\text{CropWheat} = \text{True}$

we do not know:

$\text{IrrigationNeeded} = \text{True}$

Therefore:

ApplyDripMethod cannot be derived

Step 3: Can CropAtRisk be derived?

P3 requires:

$\sim \text{ApplyDripMethod} \wedge \text{CropWheat}$

We know:

$\text{CropWheat} = \text{True}$

However, the absence of ApplyDripMethod does not mean:

$\sim \text{ApplyDripMethod}$

We only know that ApplyDripMethod cannot be proven from the remaining facts.

Therefore, we cannot infer:

$\sim \text{ApplyDripMethod}$

Hence:

CropAtRisk does not follow

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

Given Percepts

The agent perceives:

Stench at [1,2]

Breeze at [1,1]

Glitter at [2,2]

Let:

$S[1,2] = \text{Stench at } [1,2]$

$B[1,1] = \text{Breeze at } [1,1]$

$G[2,2] = \text{Glitter at } [2,2]$

$W[x,y] = \text{Wumpus at } [x,y]$

$P[x,y] = \text{Pit at } [x,y]$

$\text{Gold}[x,y] = \text{Gold at } [x,y]$

$\text{Safe}[x,y] = \text{Cell } [x,y] \text{ is safe}$

Rule 1

$S[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$

Given: $S[1,2]$

By Modus Ponens: $\text{WumpusAdjacent}[1,2]$

The cells adjacent to [1,2] are: [1,1],[1,3],[2,2]

Therefore, the Wumpus can potentially be in: [1,1],[1,3],[2,2]

Rule 2

$B[1,1] \rightarrow \text{PitAdjacent}[1,1]$

Given: $B[1,1]$

By Modus Ponens: $\text{PitAdjacent}[1,1]$

The adjacent cells are:

[1,2],[2,1]

Therefore, the pit can potentially be in:

[1,2],[2,1]

Rule 3

Glitter[x,y]→Gold[x,y]

Given: Glitter[2,2]

Substitute: x=2,y=2

Therefore: Gold[2,2]

Rule 4

$\sim \text{Wumpus}[x,y] \wedge \sim \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

To apply this rule, we need both: $\sim \text{Wumpus}[x,y]$ and $\sim \text{Pit}[x,y]$

However, the given percepts do not provide enough information to establish these negative facts for the relevant cells.

Therefore, Rule 4 cannot yet derive additional Safe facts.

Derived Belief State From the percepts and rules, the agent knows:

WumpusAdjacent[1,2]

Possible Wumpus locations: [1,1],[1,3],[2,2]

PitAdjacent[1,1]

Possible Pit locations: [1,2],[2,1] and Gold[2,2]

(b) Forward Chaining for Wumpus and Pit Locations

Forward chaining begins with the known percepts and applies rules whose antecedents are satisfied.

Step 1: Stench

Initial fact: S[1,2]

Rule:

$S[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$

Therefore: WumpusAdjacent[1,2]

Adjacent cells: [1,1],[1,3],[2,2]

So, $\text{Wumpus} \in \{[1,1],[1,3],[2,2]\}$

Step 2: Breeze

Initial fact: B[1,1]

Rule: $B[1,1] \rightarrow \text{PitAdjacent}[1,1]$

Therefore: PitAdjacent[1,1]

Adjacent cells: [1,2],[2,1]

So, Pit someof $\{[1,2],[2,1]\}$

Step 3: Glitter

Initial fact: Glitter[2,2]

Rule: $\text{Glitter}[2,2] \rightarrow \text{Gold}[2,2]$

Therefore: Gold[2,2]

Forward-Chaining Result

Stench[1,2]→ Wumpus adjacent to [1,2]→ Possible Wumpus:[1,1], [1,3], [2,2]

Breeze[1,1]→Pit adjacent to [1,1]→Possible Pit:[1,2], [2,1]

Glitter[2,2]→Gold[2,2]

Therefore:Wumpus someof $\{[1,1],[1,3],[2,2]\}$ and Pitsomeof $\{[1,2],[2,1]\}$ and Gold is at [2,2]

(c) Safety of the Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

The proposed path is:[1,1]→[1,2]→[2,2]

We must determine whether each cell is known to be safe.

Cell [1,1]

The agent starts at [1,1].

The KB contains: Breeze[1,1]

A breeze indicates that a pit may be in an adjacent cell. It does not imply that [1,1] itself contains a pit.

However, the current rules do not explicitly establish:

$\sim \text{Wumpus}[1,1]$ and: $\sim \text{Pit}[1,1]$

Therefore, according to the given KB, complete safety of [1,1] cannot formally be established using Rule 4.

Cell [1,2]

This cell is particularly dangerous.

We know: Stench[1,2] and WumpusAdjacent[1,2]

Also, from the breeze at [1,1]: Pit someof {[1,2],[2,1]}

Therefore, [1,2] is a possible pit location.

Moreover, the stench at [1,2] tells us that the Wumpus is adjacent to [1,2], but it does not mean the Wumpus is necessarily on [1,2].

Since a possible pit exists at [1,2], we cannot prove:

$\sim$ Pit[1,2]

Hence Rule 4 cannot establish: Safe[1,2]

Therefore: [1,2] is not proven safe Cell [2,2]

We know: Gold[2,2]

However, the gold does not imply that the cell is safe.

From the stench at [1,2]: Wumpus someof {[1,1],[1,3],[2,2]}

Thus:[2,2] is a possible Wumpus location

Therefore, we cannot establish: $\sim$ Wumpus[2,2] and hence cannot derive: Safe[2,2]